Franco Huidobro

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EDUCATION

Tecnológico de Monterrey Monterrey, Mexico
MSc in science of engineering (power electronics and control systems) June 2025
GPA: 4.0/4.0

Master's thesis: *A controller for an LCI to extend the operating range of single-phase brushless exciter synchronous motors.*
Advisors: Dr. Gerardo Escobar, Dr. Renato Galluzzi, M. Rommel Villarreal | **Funding:** AMI Automation

- Developed an advanced control strategy for SAG mill LCI drives, introducing a novel flux-based load-synchronized pulse scheme and real-time firing angle adjustment.
- Validated the system using HIL (dSPACE) simulations under realistic industrial milling profiles, successfully extending the motor's speed and load operating range.

Thesis published in the Tecnológico de Monterrey Institutional Repository (RITEC).

Supplementary analysis: Evaluated wireless power transfer (WPT) as an alternative excitation method at low-speed/standstill conditions via high-fidelity mathematical modeling.

Tecnológico de Monterrey Monterrey, Mexico
BSc in electromechanical engineering June 2023
GPA: 4.0/4.0 | Minor: Aeronautical design & manufacturing (GPA: 4.0/4.0)

Relevant coursework: Electric machines and electromechanical energy conversion, control and power electronics for electric machinery, industrial power systems.

PUBLICATIONS

F. Huidobro et al., "A controller for an LCI to extend the operating range of single-phase brushless exciter synchronous motors," *IEEE Access*, July 2026.

LANGUAGES

English (C1) | French (B2) | Spanish (Native)

PROFESSIONAL EXPERIENCE

AMI Automation Monterrey, Mexico
R&D power electronic control firmware developer August 2023 – Present

Specialized in the full-lifecycle R&D of control firmware for high-power thyristor and IGBT drives. The end-to-end workflow aligns firmware design with hardware capabilities, covering mathematical modeling, simulation, and real-time CPU and FPGA execution via LabVIEW. Responsibilities bridge R&D and field application by evaluating hardware-firmware dynamics through laboratory testing and leading on-site equipment commissioning.

R&D and field project: Open-winding synchronous motor cycloconverter (Nucor Steel Gallatin)
Led the R&D of a high-power cycloconverter drive for a SIEMENS rolling mill retrofit. Designed the field-oriented control (FOC) and a mixed flux observer for an open-winding synchronous motor, programming the flexible control architecture on an NI PXI system utilizing four FPGAs. Currently validating hardware-firmware dynamics through closed-loop laboratory testing with physical machines and a 6-pulse cycloconverter, ahead of the April 2027 on-site commissioning.

R&D and field project: Electric arc furnace retrofit (Nucor Steel Darlington)
Simulated a 12-pulse rectifier utilizing a Chua's equations-based chaotic model to calculate the optimal interphase reactor. It minimized voltage losses and increases power transfer to the arc for faster melting cycles. Validated current control through laboratory testing with RL loads and current step responses, concluding with the on-site commissioning in March 2027.

R&D and field project: Static var compensator (SVC) integration (Rocky Mountain Steel)
Designed and programmed real-time power quality algorithms for the AMI SmartDisplay HMI® to quantify SVC performance. Implemented measurements including flicker evaluation, voltage and current harmonic analysis, and real-time short-circuit power estimation, finishing in the successful commissioning in November 2026.

R&D and field project: Electromagnetic stirring (EMS) IGBT drive (Rocky Mountain Steel)
Designed an advanced IGBT inverter control architecture for a magnetohydrodynamic machine to optimize the electromagnetic stirring of molten steel. Reverse-engineered the existing installation to formulate a new stirring profile based on the applied Lorentz forces which translated into current references. Programmed the control logic, and validated the system performance, with on-site commissioning in February 2027.

R&D and field project: Power quality & energy storage (Resonac Graphite Japan)
Engineered a supercapacitor-based solution for a DC bus topology to protect critical loads from voltage dips. Modeled the full rectifier-inverter and grid interaction dynamics and calculated optimal supercapacitor sizing. Validated current control logic for capacitor charging and translated the final control laws to NI targets Successful deployment in May 2026.

R&D and field project: DigiFex® field excitation system upgrades (CEMEX)
Led the control firmware architecture updates for an across-the-line synchronous machine excitation drive, responsible for starting sequences and PF/VAR regulation. Engineered two major updates:

- **Digital twin integration:** Programmed a mathematical-model-based digital twin of the synchronous machine, enabling hardware-free simulations of control loops, logic, I/O, and protections to accelerate firmware validation and facilitate technical demonstrations.
- **Machine learning fault detection:** Programmed a Support Vector Machine (SVM) algorithm to detect thyristor misfiring. Trained offline via MATLAB/Simulink, the model analyzes rectified voltage to identify firing failures, preventing AC-side unbalances. First on-site commissioning is scheduled at CEMEX, standardizing the feature for future units with planned expansion to cycloconverters.

R&D and field project: Supersynchronous induction drive (Brookhaven Laboratory)
Design and implementation of FOC for a wound-rotor 2.4 Hz induction motor slip power recovery drive. To meet laboratory requirements for the machine's supply current waveform (THD $< 1\%$ and current ripple $< 0.1\%$), a 12-pulse non-circulating current cycloconverter was proposed. Optimal sizing for both the interphase and series reactors was determined through simulation to successfully achieve these power quality criteria.

R&D and field project: Model predictive control (MPC) for electric arc furnaces
Developed an advanced MPC algorithm for a high-speed rectifier to mitigate grid flickering and regulate

DC current under chaotic load dynamics typical of steel processes. Validated the controller's robustness against extreme load variations via Simulink simulations.

Field project: Quadra-torque drive system commissioning (Minera Peñasquito)
Led the end-to-end commissioning of four twin-synchronous motor ball mill drives, ensuring seamless integration and supervising technician teams to minimize downtime in September 2025. Designed and delivered an advanced technical training curriculum for 70 site engineers, covering drive theory, fault analysis, and operational safety.

Tecnológico de Monterrey Monterrey, Mexico
Graduate teaching assistant & laboratory instructor August 2023 – June 2025

Power electronics laboratory & program coordination
Instructed laboratory sessions focused on DC-DC and DC-AC converters, supervising students in practical circuit analysis, hardware debugging, and rigorous safety protocols. Additionally, collaborated with the undergraduate program director (Dr. Esmeralda Uribe Lam) to evaluate performance in aeronautical and materials engineering coursework, and contributed to the redesign of curricula.

Tecnológico de Monterrey Monterrey, Mexico
Research assistant November 2022 – August 2023

Embedded control systems
Designed and implemented a opened-loop AC motor speed control system programmed in C utilizing a Microchip dsPIC, successfully integrating a three-phase IGBT inverter. The prototype was deployed as an active educational module for undergraduate courses, and the technical design was presented at the university science showcase "Expo-Ingenierías".

Tecnológico de Monterrey / Lean Methods Group Monterrey, Mexico
Lean Six Sigma advisor August 2021 – July 2022

Process optimization & statistical analysis
Co-led manufacturing optimization projects for ceramic and tantalum capacitors, utilizing advanced statistical analysis in Minitab (ANOVA, DOE, Gage R&R) to achieve $660k USD in annual savings. Concurrently earned the Lean Six Sigma Black Belt and Design for Six Sigma (DFSS) Green Belt certifications, successfully demonstrating strong project management and analytical capabilities.

TECHNICAL TRAINING & ACADEMIC OUTREACH

AMI TechTalk: Synchronous motors technology and applications Houston, USA
Technical instructor March 2025, March 2026, July 2026

Led three technical training courses for industrial clients, combining drive theory with product demonstrations. Taught modules on synchronous machine fundamentals, torque production, across-the-line starting, brushed/brushless excitation, and field-oriented control (FOC). Developed and presented custom simulations to demonstrate control algorithms and hardware capabilities.

AMI-Tec de Monterrey industry partnership (Socio Formador) Monterrey, Mexico
Industrial mentor Spring 2026

Mentored undergraduate electronics engineering students in a joint industry-academia project to power water pumps in off-grid areas. Defined the technical scope and control theory three-phase inverter and induction motor. Guided students through hardware implementation and debugging during lab sessions, and evaluated the final prototypes as a jury member.

HONORS, AWARDS & AFFILIATIONS

Professional affiliations: Member of the Institute of Electrical and Electronics Engineers (IEEE).

Honorable mention of excellence (summa cum laude) 2023
Highest academic distinction awarded by Tecnológico de Monterrey to the top 2.5% of the graduating class for exceptional GPA and academic integrity.

National award CENEVAL 2023
Awarded by the National Evaluation Center for Higher Education to the top 1.5% nationwide for outstanding performance in the General Exit Examination for Bachelor's Degrees (EGEL).

Leadership in professional development LIFE award 2023
Distinguished for demonstrating exceptional leadership and problem-solving capabilities during professional internships and industrial placements.

Outstanding trajectory of the school of engineering and sciences 2022
Recognized for maintaining a balance of excellence across academic research, industrial placements, and extracurricular leadership during the undergraduate program.

INTERESTS & SKILLS

Research interests: Control and new topologies for thyristors-based drives and electrical machines. Strong focus on high-power thyristor-based converters (cycloconverters, LCIs, rectifiers, SVCs) and their advanced control applications for heavy steel industry processes. Highly motivated to leverage this background in nonlinear control (model predictive control, sliding mode control) and state estimation to tackle emerging challenges in high-voltage direct current (HVDC) systems, solid-state transformers, and next-generation high-power topologies.

Modeling & simulation: MATLAB, Simulink, Stateflow, mathematical modeling (differential equations), SPICE.

Embedded control & FPGA: LabVIEW Real-Time, NI FPGA module, NI cRIO, Microchip dsPIC, C/C++ programming for embedded systems.

Hardware & testing: Hardware-in-the-loop (HIL, dSPACE), rapid control prototyping (RCP), power analyzers, electronic instrumentation, industrial commissioning.

Analysis & quality: Minitab (DOE, ANOVA, hypothesis testing).